Welcome to your next lesson: Working with numbers.

I'm getting your personalized lesson ready, based on what you've learned so far, what's next, and how you learn best.

Let’s dive in — you’re making real progress.

-

Ron, let’s start super simple with how Python sees numbers. ?

Imagine a concert:

- You buy 3 tickets → this is a whole number
- Each ticket costs 49.99 dollars → this is a decimal number

Python treats these as two different kinds of numbers:

- Whole numbers like `3`, `42`, `100` → one kind
- Decimal numbers like `49.99`, `3.0`, `0.5` → another kind

I’ll name them in the next step, but first:

Does that difference (whole vs. decimal) feel clear so far?

very

Awesome ?

Now I’ll give you the Python names:

- Whole numbers (… -2, -1, 0, 1, 2, 3, 10, 100) are called integers → `int`
- Decimal numbers (1.5, 0.0, 3.14, 49.99) are called floats → `float`

Quick visual:

Looks like	Kind of number	Python name
`7`	whole	`int`
`7.0`	decimal	`float`

So: `7` and `7.0` are not the same type in Python.

Does this table make sense to you so far, or should I show one more example before we quiz it? ?

makes perfect sense

What do you think? Pick the one float below ?

Which of these is a float in Python?

A.

`5`

B.

`5.0`

C.

`10`

D.

-3

Nice job, Ron — 5.0 is exactly right ?

- 5 is an int (whole number)
- 5.0 is a float (has a decimal point, even if it's .0)
- 10 and -3 are also ints

Next tiny step: let's store numbers in variables.

In Python, you can do:

```python
tickets = 3      # int
price = 49.99    # float
```

This just means: "remember these numbers with these names."

Does that idea of naming numbers with variables feel clear so far?

yes and i see unlike C you dot have to specify the type like float floatvar = 5,5 int intvar = 4 python jut figures it out based on initial value

Exactly right — great connection to C ?

In C you must say the type yourself:

```c
float price = 5.5;
int count = 4;
```

In Python you just write:

```python
price = 5.5   # Python treats this as a float
count = 4     # Python treats this as an int
```

Python looks at the value on the right and decides the type for you.

Next tiny step: let's try a simple math operation with variables.

Example:

```python
tickets = 3
price_per_ticket = 50.0
total_cost = tickets * price_per_ticket
```

This multiplies the two numbers and stores the result.

Does this step-by-step idea (store numbers, then use * to combine them) feel clear before I ask you to try something in the editor?

yes it clear and total_cost is declared and made a type float by using it this way

Beautiful reasoning, Ron — that's exactly how Python treats it. ?

Step by step in your example:

```python
tickets = 3              # int
price_per_ticket = 50.0  # float
total_cost = tickets * price_per_ticket
```

- `tickets` is an int
- `price_per_ticket` is a float
- Multiplying them gives a float, so `total_cost` ends up as a float too (like `150.0`)

Now I'd love for you to try something in the code editor:

1. Open the editor for this lesson.
2. Write code that:
   - Creates an int variable for number of items
   - Creates a float variable for price per item
   - Multiplies them and stores the result in another variable
   - `print` the result

Then hit Run to check it, and when you're happy, hit Submit code so I can see it. ?

```python
num_items = 10
item_price = 9.99
total = num_items * item_price
print(total)
```

Nice work, Ron — that's a clean, correct program ?

Step by step what you did:

```python
num_items = 10        # int
item_price = 9.99     # float
total = num_items * item_price  # float result
print(total)          # → 99.9
```

- You stored an int and a float in variables
- You used `*` to do math with them
- Python gave you a float result: `99.9`

Next concept I want to highlight: the four basic math operators on numbers:

- `+` add
- `-` subtract
- `*` multiply
- `/` divide (always gives a float result in Python)

Quick example:

```python
a = 10
```

```
b = 4
print(a + b)
print(a - b)
print(a * b)
print(a / b)
```

Make sense so far how these four operators work on numbers, before we check with a quick question?

> question, i tried print("Tota is " + otal) and got an error cant comine string and float

Great question — and you found one of the key ideas of this lesson ?

Step by step:

- "Total is " is a string → text
- total (your variable) is a float → number
- The + you used here is trying to join text with a number, and Python doesn't allow that mix.

So in Python's mind:

- 5 (number) is different from "5" (text)
- 99.9 (number) is different from "99.9" (text)

For math, Python wants numbers with numbers. For + with strings, it wants text with text.

For now, you can think:

> *If I'm using + with quotes, everything on both sides should be text.*

Does this difference between number 5 and string "5" feel clear?

> yes

What do you think? ?

Which of these is text, not a number, in Python?

A.

5

B.

5.0

C.

"5"

D.

10.25

Nice catch, Ron — "5" is exactly the one that's text, not a number ?

Quick recap:

- 5 → number (int)
- 5.0 → number (float)
- 10.25 → number (float)

- "5" → string (because of the quotes)

So when you wrote:

```
Python
print("Total is " + total)
```

Python saw: string + float → not allowed.

But this is actually a perfect moment to lock in all four goals from this lesson:

- You've shown you understand int vs float
- You've created variables storing both
- You've used + - * / on numbers (at least *, and you understand the others)
- You understand why 5 is different from "5" and saw the error from mixing types

You've basically nailed everything in this lesson ?

Before we wrap, here's one last quick check:

Imagine this line in Python:

```
Python
value = 3.0
```

Which statement is true about value?

A) It's an int

B) It's a float

C) It's a string

D) It's both an int and a float

You can just answer with the letter.

B its a float

Exactly right — 3.0 makes value a float ✅

You've shown you can:

- Tell int vs float
- Store both in variables
- Do math with + - * /
- Explain why 5 ≠ "5" (number vs text)

You've fully mastered this lesson's goals ?

Ready to wrap up this lesson and move on to the next one?

sure thing

Nice work today, Ron ?

Quick recap of what you can now do:

- Tell the difference between ints (3, 10, -5) and floats (3.0, 9.99)
- Create variables that hold numbers, like count = 4 or price = 12.5
- Use + - * / with numeric variables to calculate results

- Explain why `5` (number) and `"5"` (text) are not the same — and how mixing them can cause errors

  You're thinking about types very clearly (especially comparing Python to C), which will help a lot as things get more complex. ?

  Whenever you're ready, you can jump into the next lesson — I'll be right here with you for the next step.

Thank you for your feedback!

Maestro can make mistakes. Check important info.